F-Fe-MOF Functionalized PVDF-HFP Nanofibrous Mats-based Triboelectric Nanogenerator for Self-Powered Respiratory Rehabilitation Monitoring

Md Shofiul Alam, M. Robiul Islam, Moon Seong Jo, Jae Yeong Park*
Kwangwoon University, Department of Electronic Engineering.
E-mail: jaepark@kw.ac.kr

자가 전원 호흡기 재활 모니터링을 위한 F-Fe-MOF 로 기능화된 PVDF-HFP 나노섬유 매트 기반 마찰전기 나노발전기

알람쇼피울, M. 로비울 이슬람, 조문성, 박재영*
광운대학교 전자공학과
E-mail: jaepark@kw.ac.kr

Abstract

Herein, we report a highly sensitive self-powered wearable pressure sensor (SWPS) for human respiratory rehabilitation monitoring using a high-performance triboelectric nanogenerator comprised of fluorinated Fe-MOF–functionalized PVDF-HFP nanofibrous mat (F-FeM@PNM) as a strongly negative triboelectric layer and PEO nanofibers as the tribo-positive counterpart. The incorporation of fluorinated Fe-MOF (F-Fe-MOF) into the PVDF-HFP matrix induces a pronounced enhancement in electroactive β-phase crystallinity, which synergistically amplifies the dielectric properties, interfacial polarization, and surface charge density, thereby unlocking the triboelectric output performance. Moreover, the abundant charge-trapping sites and intrinsic nano-porosity of nanofillers facilitate deep penetration into the polymer matrix, enhancing electrostatic induction and charge-retention ability, and ensuring high energy conversion efficiency. The developed SWPS exhibited an ultrahigh sensitivity of 5.41 $VkPa^{-1}$ in the low-pressure regime (0.25–5 kPa) and delivers a peak power density of 2.21 Wm^{-2}. It was successfully demonstrated as a real-time respiration rehabilitation monitoring system via a custom-developed computer app interface.

Keywords: Fluorinated Fe-MOF, F-Fe-MOF@PVDF-HFP nanofibrous mats, triboelectric nanogenerator, wearable self-powered respiration rehabilitation system.

1. Introduction

Substantial progress has been made in triboelectric materials for self-powered sensing systems; however, their practical application in low-pressure wearable sensing such as human activity and respiration rehabilitation monitoring remains limited by the intrinsically low dielectric constant, insufficient charge retention, and weak interfacial polarization of conventional polymers [1,2]. These limitations lead to low output performance and poor signal stability under subtle biomechanical stimuli. Moreover, most existing wearable triboelectric sensors suffer from inadequate long-term stability, compromised wearing comfort, and low sensitivity-related issues, which restrict their use in real-world scenarios such as continuous respiration monitoring [3]. To address these challenges, we report the rational design of a fluorinated Fe-MOF functionalized PVDF-HFP nanofibrous triboelectric material (F-FeM@PNM) and its integration into a multifunctional self-powered smart chest band for reliable and continuous respiration rehabilitation monitoring.

2. Experimental Results

Figure 1a shows the molecular structure and fabrication route of the fluorinated Fe-MOF@PVDF-HFP nanofibrous mat (F-FeM@PNM). FTIR spectroscopy (Figure 1b) confirms the successful fluorination, and FESEM images (Figures 1c-d) visualize the porous nanofillers and their homogeneous dispersion within the fibrous matrix. The porous fluorinated Fe-MOF functions as an effective nucleating agent, facilitating uniaxial nanocrystal alignment and stabilizing the β-phase, as shown in Figures 2a and 2b. The incorporation of fluorinated Fe-MOF (0–20 wt.%) into PVDF-HFP matrix improves the dielectric constant and output voltage, with an optimum at 10 wt.% (Figure 3a). The F-FeM@PNM exhibited a maximum transferred charge of 38.2 nC and a surface potential of −4.81 kV, both substantially higher than those of pristine PVDF-HFP nanofibers (Figure 3b). Superior charge-retention efficiency was demonstrated by a reduced decay rate of 43.48% over 10 h, compared to 98.5% for PVDF-HFP nanofibers (Figure 3c). Under a frequency of 3 Hz, the device exhibited a peak power density of 2.21 W m^{-2} (Figure 3d). The SWPS exhibited a high-pressure sensitivity of 5.41 V kPa^{-1} over a range of 0.25–5 kPa (Figure 4a). Finally, the SWPS was attached with a chest band to monitor the human respiratory rehabilitation through commercial computer app interface shown in Figure 4b. It was highly effective to distinguish diverse respiration patterns (normal, rapid, and apnea) (Figure 5c–d).

3. Conclusion

In this study, we successfully developed high-performance triboelectric nanogenerators (TENGs) based on nanofibrous mats incorporating fluorinated Fe-MOFs. The

introduction of fluorinated Fe-MOFs induced a β-phase–enriched structure and enhanced charge-trapping capability in the F-FeM@PNM, enabling self-powered wearable respiration monitoring. The electronegative fluorinated Fe-MOF significantly increased the β-phase content to 98.7%, thereby enhancing interfacial polarization and charge retention. At an optimal loading of 10 wt.%, the fabricated TENG exhibited excellent triboelectric performance, delivering an output voltage of approximately 684 V, a transferred charge of 36.4 nC, and a power density of 2.1 W m^{-2}. It also demonstrated outstanding operational stability and was capable of directly powering small electronic devices. In addition, it functioned as a highly sensitive pressure sensor, with a sensitivity of 5.41 V kPa^{-1} over a pressure range of 0.25–5 kPa. This dual-functional device enables efficient energy harvesting as well as real-time monitoring of respiration and rehabilitation progress, offering a promising platform for intelligent and self-powered rehabilitation applications.

Acknowledgments

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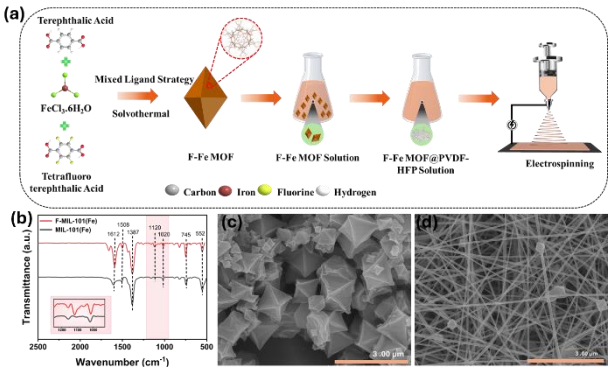

Figure 1: (a) Schematic representation of the fabrication process of fluorinated Fe-MOF, F-Fe-MOF@PVDF-HFP nanofibers. (b) FT-IR spectrum of Fe-MOF and fluorinated-Fe-MOF. The surface morphology of (c) fluorinated-Fe-MOF, (d) F-Fe-MOF@PVDF-HFP analyzed via FESEM.

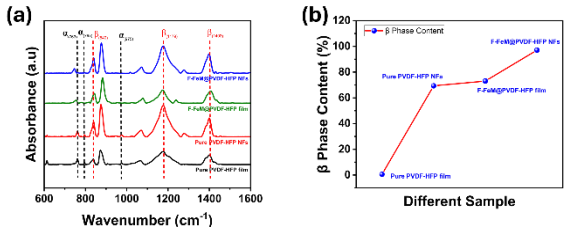

Figure 2: (a) FT-IR spectrum analysis (b) Calculated polar β-phase content of different pure PVDF-HFP variants and 10 wt.% F-FeM@PNM Variants.

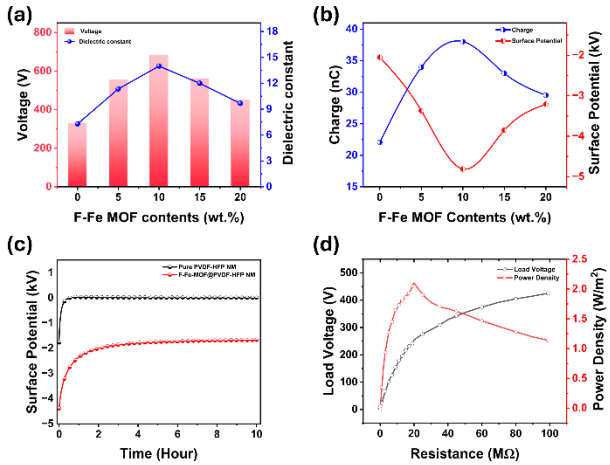

Figure 3: (a) The output voltage and dielectric constant of the TENG at varying concentrations of F-Fe-MOF. (b) Transferred charge and surface potential of the TENG at different concentrations of F-Fe-MOF. (c) The charge retention capability of pristine PVDF-HFP and F-FeM@PNM. (d) The output performance of the TENG at various external load resistances.

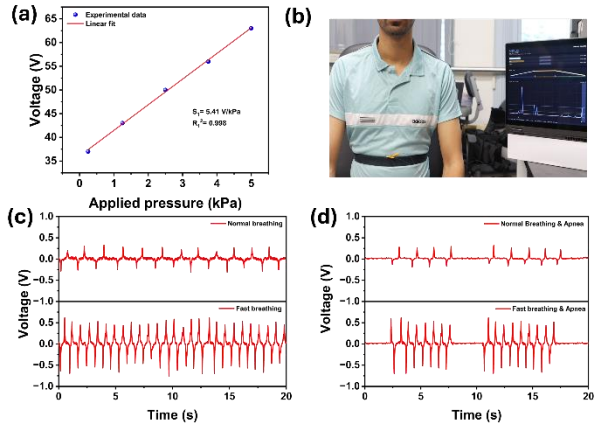

Figure 5: (a) Pressure sensitivity of the SWPS at low pressure regime (0-5 kPa). (b) Real-time photographs of respiratory rehabilitation progress using custom-made computer app interface. (c-d) Detection of different breathing patterns such as continuous Normal, Fast, Apnea using SWPS on chest.